

Coordinate-Wise Robustness of Neural SDE Forecasts under Missingness and Input Noise

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Abstract

Neural stochastic differential equations (SDEs) provide a continuous-time model for partially observed dynamics while representing stochasticity explicitly. Their apparent robustness, however, can depend strongly on the state coordinate and on the scale at which prediction differences are measured. We compare a neural controlled differential equation (NCDE) forecaster with four neural SDE variants—LSDE, LNSDE, GSDE, and NaiveSDE—on 14-dimensional MuJoCo Hopper trajectories. Each SDE prediction is estimated from 20 Monte Carlo paths. The trained models are evaluated, without retraining, under 12 combinations of missing rate and additive input noise, with five independently generated corruption patterns per condition.

The raw ODE–SDE discrepancy is concentrated in several coordinates, most visibly channels 4, 5, and 11. Scale normalization reveals a different pattern: channel 10 (waist angular velocity) has the second-largest coordinate-wise discrepancy, reaching 3.30–5.74 target standard deviations across SDE architectures, compared with 5.04–5.56 for channel 11. The mechanisms differ. For channel 10, 97.2–99.8% of squared ODE–SDE disagreement is a sign-changing component across examples, forecast horizons, and corruption conditions; for channel 11, 69.1–83.0% is explained by a persistent positive offset. Increasing input noise from 0 to 0.2 raises the channel-10 discrepancy by 2.05–5.22%, whereas increasing missingness from 0 to 0.7 changes it by at most 1.10% in magnitude. These results identify channel 10 as a scale-sensitive candidate for further stochastic analysis, but they do not by themselves establish chaos or measure coordinate-wise Monte Carlo variance.

1 Introduction

Neural ordinary differential equations (ODEs) parameterize continuous-time latent dynamics with a neural vector field [1]. Neural controlled differential equations (CDEs) extend this framework by driving the latent state with an interpolated observation path and are therefore well suited to irregular and partially observed time series [2]. Neural SDEs add a diffusion term, allowing the model to represent stochastic dynamics and predictive uncertainty. This additional flexibility also introduces new questions: stability of the result and influence of diffusion and drift function on our final result.

Aggregate forecasting loss alone cannot answer these questions. A model may have a small global mean-squared error while exhibiting a large ODE–SDE discrepancy in a coordinate with a small natural scale. Conversely, a large raw discrepancy can merely reflect a high-variance coordinate. We therefore study both raw and scale-normalized coordinate-wise differences between an NCDE forecast and the Monte Carlo mean of a neural SDE forecast.

Our contributions are threefold:

- We evaluate four neural SDE parameterizations under a paired 4×3 grid of missingness and additive input noise, without retraining inside the stress-test grid.
- We separate global predictive variance from coordinate-wise ODE–SDE model disagreement and correct the interpretation of the SDE expectation.
- We show that channel 10 is modest in raw units but is one of the largest discrepancies relative to its target scale, and that its disagreement is primarily sign-changing rather than a persistent bias.

2 Related Work

Neural ODEs introduced black-box differential-equation solvers as trainable continuous-depth models [1]. Latent ODEs subsequently adapted continuous-time latent-variable models to irregularly sampled observations [3]. Neural CDEs provide a principled mechanism for continuously updating a latent state from an observation path and have become a standard baseline for irregular time-series modelling [2].

Neural SDEs augment the drift with a learned diffusion function. Stable Neural SDEs constrain drift and diffusion parameterizations to improve well-posedness and numerical behaviour on irregular time series [4]. Our objective is not to introduce another SDE architecture. Instead, we analyze how existing SDE forms disagree with a deterministic NCDE forecaster under controlled observation corruption, with particular attention to physical coordinates that are hidden by aggregate metrics.

3 Experimental Setup

3.1 Dataset and forecasting protocol

We use the 14-dimensional MuJoCo Hopper trajectories distributed with the forecasting benchmark. The first seven channels are generalized positions and the final seven are their corresponding generalized velocities. In the joint ordering used by the Hopper model, channel 10 is the waist angular velocity and channel 11 is the hip angular velocity. Each trajectory is cut into overlapping windows with 50 input points and a 20-step forecasting target. This produces 3,000 windows, split into 2,100 training, 450 validation, and 450 test examples.

The training inputs use a missing rate of 0.3. Each ODE or SDE model is trained once. At test time, the fixed model weights are evaluated on missing rates

$$q \in \{0.0, 0.3, 0.5, 0.7\}$$

and additive input-noise levels

$$\eta \in \{0.0, 0.1, 0.2\}.$$

Only the observed input portion is corrupted; forecasting targets remain unchanged. Five corruption seeds are used per condition. These repeats generate different missing-data and input-noise patterns; they are distinct from the 20 Brownian paths used for the SDE Monte Carlo estimate.

3.2 Models

The deterministic baseline is an NCDE forecaster. We compare it with LSDE, LNSDE, GSDE, and NaiveSDE variants from the Stable Neural SDE benchmark [4]. All models use the same data split, input window, forecast horizon, and training seed. For every test example and corruption condition, the SDE point forecast is the empirical mean of $M = 20$ valid, non-exploded paths. Reported SDE predictive variance is computed over those paths. The stress grid reuses trained weights; no model is retrained for a particular missing rate or noise level.

4 Evaluation Metrics

Let y_{nhj} be the target for example n , forecast step h , and coordinate j ; let o_{nhj} be the ODE prediction; and let $s_{nhj}^{(m)}$ be SDE path m . The Monte Carlo mean is

$$\bar{s}_{nhj} = \frac{1}{M} \sum_{m=1}^M s_{nhj}^{(m)}. \quad (1)$$

With $D = 14$ output coordinates, the ODE and SDE-mean losses are

$$L_O = \frac{1}{NHD} \sum_{n,h,j} (o_{nhj} - y_{nhj})^2, \quad (2)$$

$$L_S = \frac{1}{NHD} \sum_{n,h,j} (\bar{s}_{nhj} - y_{nhj})^2. \quad (3)$$

Thus the corresponding root-mean-square distances to the target are $R_{OY} = \sqrt{L_O}$ and $R_{SY} = \sqrt{L_S}$.

We measure the global ODE–SDE discrepancy as

$$R_{OS} = \sqrt{\frac{1}{NHD} \sum_{n,h,j} (\bar{s}_{nhj} - o_{nhj})^2}, \quad (4)$$

and its coordinate-wise counterpart as

$$R_{OS,j} = \sqrt{\frac{1}{NH} \sum_{n,h} (\bar{s}_{nhj} - o_{nhj})^2}. \quad (5)$$

Because the channels have different scales, we additionally report

$$\tilde{R}_{OS,j} = \frac{R_{OS,j}}{\sigma_{y,j}}, \quad (6)$$

where $\sigma_{y,j}$ is the population standard deviation of coordinate j over the fixed test targets. This normalization is a diagnostic used to compare coordinates; it is not applied to the training objective or to L_O and L_S .

The aggregate SDE predictive variance is

$$V_S = \frac{1}{NHD} \sum_{n,h,j} \left[\frac{1}{M-1} \sum_{m=1}^M \left(s_{nhj}^{(m)} - \bar{s}_{nhj} \right)^2 \right]. \quad (7)$$

To retain the output-horizon structure, we additionally report

$$V_S(h) = \frac{1}{ND} \sum_{n,j} \left[\frac{1}{M-1} \sum_{m=1}^M \left(s_{nhj}^{(m)} - \bar{s}_{nhj} \right)^2 \right]. \quad (8)$$

The pooled horizon profile averages $V_S(h)$ over the 12 equally weighted corruption conditions, each already aggregated over five corruption realizations. This quantity describes Monte Carlo forecast dispersion at each output position; it is not an uncertainty-calibration score. Finally, loss degradation under a corruption condition c is measured relative to the reference view $(q, \eta) = (0.3, 0)$:

$$\delta_O^{(c)} = \frac{L_O^{(c)}}{L_O^{(\text{ref})}} - 1, \quad \delta_S^{(c)} = \frac{L_S^{(c)}}{L_S^{(\text{ref})}} - 1. \quad (9)$$

4.1 What the SDE mean does and does not represent

For the Itô SDE

$$dZ_t = f_\theta(Z_t, t) dt + g_\phi(Z_t, t) dW_t, \quad (10)$$

standard integrability assumptions imply

$$\mathbb{E}[Z_t] = \mathbb{E}[Z_0] + \int_0^t \mathbb{E}[f_\theta(Z_\tau, \tau)] d\tau, \quad (11)$$

because the Itô integral has zero expectation. In general,

$$\mathbb{E}[f_\theta(Z_t, t)] \neq f_\theta(\mathbb{E}[Z_t], t), \quad (12)$$

so the diffusion can affect the mean indirectly by changing the distribution at which the nonlinear drift is evaluated. Moreover, the NCDE and SDE models are trained separately and do not share a drift network. Consequently, R_{OS} quantifies model disagreement; hence helping us to see their drift disagreement

Table 1: Pooled performance and coordinate-wise NCDE–SDE discrepancy across the 12 stress-test conditions. Standardized values use the population standard deviation of the fixed test targets.

Model	L_S	V_S	R_{OS}	$R_{OS,10}$	$\tilde{R}_{OS,10}$	$R_{OS,11}$	$\tilde{R}_{OS,11}$
LSDE	0.04399	7.626e-04	0.20341	0.08593	4.11	0.45392	5.37
LNSDE	0.05660	9.980e-04	0.20120	0.11293	5.40	0.42635	5.04
GSDE	0.04538	1.788e-04	0.20131	0.11997	5.74	0.46989	5.56
NaiveSDE	0.06217	1.779e-04	0.18911	0.06909	3.30	0.44544	5.27

Table 2: SDE predictive variance at the first and final forecast horizons, pooled over the 12 corruption conditions and five repeats. The ratio describes net horizon-wise growth; it is not a stability or chaos criterion.

Model	$V_S(0)$	$V_S(19)$	$V_S(19)/V_S(0)$	Horizon profile
LSDE	5.486e-04	9.496e-04	1.73	monotone
LNSDE	5.308e-04	1.611e-03	3.04	monotone
GSDE	1.623e-04	1.988e-04	1.23	non-monotone
NaiveSDE	1.241e-04	3.623e-04	2.92	monotone

5 Results

5.1 Aggregate forecasting and predictive variance

Across the 12 corruption conditions, the common ODE baseline has pooled $L_O = 0.07841$. LSDE achieves the smallest SDE-mean loss ($L_S = 0.04399$), while GSDE and NaiveSDE have the smallest aggregate predictive variances (1.788×10^{-4} and 1.779×10^{-4} , respectively). The global ODE–SDE discrepancy remains similar across architectures, from 0.18911 for NaiveSDE to 0.20341 for LSDE. Table 1 summarizes the aggregate results; the complete 12-condition tables are moved to Appendix A.

5.2 Horizon-wise predictive variance

The aggregate predictive variance conceals a pronounced architecture-dependent horizon profile. After pooling the 12 corruption conditions, LSDE increases from 5.49×10^{-4} at $h = 0$ to 9.50×10^{-4} at $h = 19$, corresponding to a factor of 1.73. LNSDE shows the strongest increase, from 5.31×10^{-4} to 1.61×10^{-3} ($3.04\times$). NaiveSDE starts at the lowest level, 1.24×10^{-4} , but reaches 3.62×10^{-4} ($2.92\times$), with most of the acceleration occurring at the final output positions. GSDE has the weakest net increase, from 1.62×10^{-4} to 1.99×10^{-4} ($1.23\times$), and its profile is non-monotone.

The endpoint increase is positive under every tested corruption condition for every architecture, while condition-to-condition changes in the ratios are small. Horizon-wise variance therefore separates the SDE forms more clearly than the nearly unchanged aggregate corruption losses in this experiment. Absolute variance and relative growth must be considered together: NaiveSDE grows almost as much as LNSDE proportionally but remains substantially lower in absolute variance. Lower variance is not automatically preferable, as it may not match variance of the law of input data, however from Stable NSDE point of view it showing steadiness to the missing data factor.

5.3 Coordinate-wise discrepancy

Figure 2 shows that raw and scale-normalized analyses lead to different conclusions. In raw units, channels 4, 5, and 11 dominate. After dividing by the target standard deviation, channel 10 becomes the second-largest coordinate on average across architectures. Its pooled $\tilde{R}_{OS,10}$ is 4.11 for LSDE, 5.40 for LNSDE, 5.74 for GSDE, and 3.30 for NaiveSDE. Channel 11 remains large at 5.04–5.56. The corresponding target standard deviations are $\sigma_{y,10} = 0.0209173$ and $\sigma_{y,11} = 0.0845872$.

The physical interpretation is also different: channel 10 is waist angular velocity, whereas channel 11 is hip angular velocity. A large normalized value therefore identifies a coordinate for targeted follow-up; it does

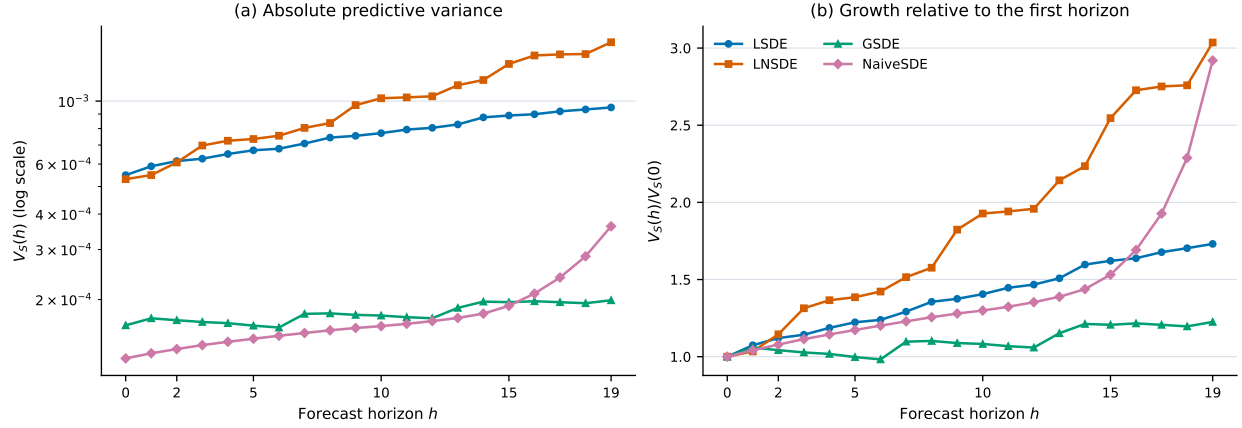


Figure 1: SDE predictive variance across output forecast horizons, pooled over the 12 corruption conditions. Left: absolute variance on a logarithmic scale. Right: variance relative to the first output horizon. Shaded regions show the minimum–maximum range of condition means and are not confidence intervals. These profiles describe architecture-specific forecast dispersion; they do not establish calibration, chaos, however showing stability to the different noise

not prove that the underlying joint is chaotic. Establishing chaos would require a separate sensitivity analysis, for example through Lyapunov exponents or controlled perturbation growth.

5.4 Why channels 10 and 11 differ

Define $\Delta_j = \bar{s}_j - o_j$ and decompose the squared RMS discrepancy as

$$R_{OS,j}^2 = \mathbb{E}[\Delta_j]^2 + \text{Var}(\Delta_j). \quad (13)$$

For channel 10, only 0.23–2.75% of $R_{OS,10}^2$ is explained by a systematic offset; 97.25–99.77% is the varying component across examples, horizons, and corruption conditions. Channel 11 shows the opposite pattern: a positive systematic offset explains 69.08–83.01% of its squared discrepancy. Table ?? gives the channel-10 values, and Figure 3 contrasts the two coordinates.

5.5 Sensitivity to missingness and input noise

After pooling over missing rates, raising input noise from 0 to 0.2 increases $R_{OS,10}$ by 3.51% for LSDE, 2.05% for LNSDE, 2.58% for GSDE, and 5.22% for NaiveSDE. After pooling over noise levels, raising the missing rate from 0 to 0.7 changes the same metric by -0.75% , -0.51% , -1.10% , and $+0.62\%$, respectively. Within this grid, channel 10 is therefore more responsive to additive input noise than to random missingness. Channel 11 has a smaller noise effect (0.57–0.81%) but a consistent missingness effect (1.45–1.60%). The full channel-10 condition table and heatmaps appear in Appendix B.

6 Limitations

First, the experiment uses one training seed and one model instance per architecture. The five repeats vary only the test corruption pattern, so they do not quantify training uncertainty. Second, the architectures do not have identical parameter counts, and raw MSE emphasizes coordinates with large numerical scale. Third, the current result files store V_S only after averaging over all 14 coordinates. They do not store coordinate-wise Monte Carlo variance $V_{S,j}$ or coordinate-wise loss to truth. Accordingly, the variation of Δ_{10} reported here must not be interpreted as stochastic path variance. A follow-up experiment should record $L_{O,j}$, $L_{S,j}$, and

$$V_{S,j} = \frac{1}{NH} \sum_{n,h} \frac{1}{M-1} \sum_m \left(s_{nhj}^{(m)} - \bar{s}_{nhj} \right)^2 \quad (14)$$

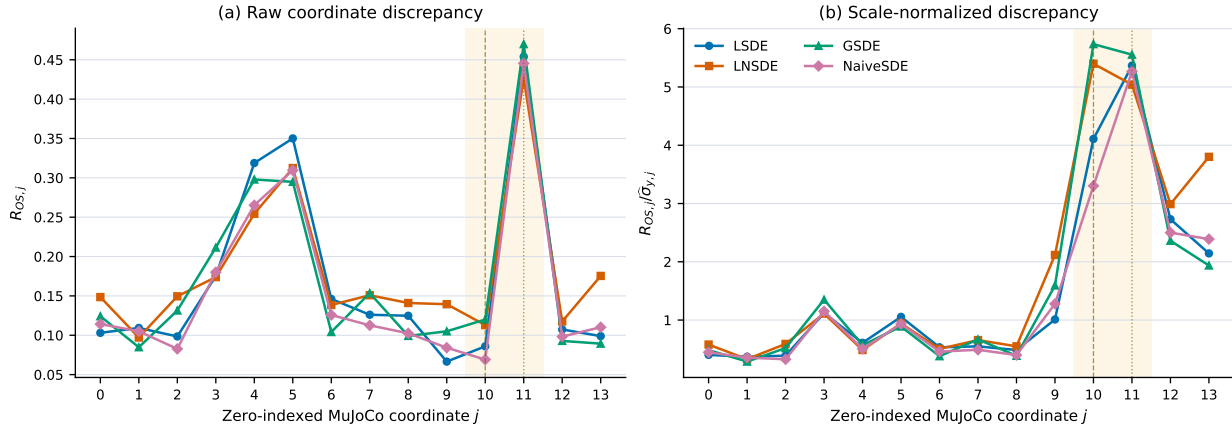


Figure 2: Coordinate-wise ODE-SDE disagreement pooled over the 12 stress-test conditions. Left: raw RMS discrepancy. Right: discrepancy normalized by the population standard deviation of the fixed test targets. Scale normalization reveals channel 10 as the second-largest relative discrepancy. The plotted quantity is model disagreement, not Monte Carlo predictive variance.

for every coordinate. Finally, the scale normalization uses the fixed test-target distribution and is descriptive; future model selection should use scales estimated from training data only.

7 Conclusion

The aggregate losses suggest that all four neural SDE variants remain well behaved under the tested observation corruptions, but coordinate-wise analysis reveals structure that the global averages conceal. Channel 10 has a relatively small raw discrepancy yet a large discrepancy relative to its natural target scale. Its sign-changing behaviour differs qualitatively from the persistent positive offset observed for channel 11, and it responds more strongly to additive input noise than to missingness. These findings make channels 10 and 11 useful targets for comparing different diffusion function forms and more detailed analysis of NSDE. They do not, without coordinate-wise path variance and a dynamical sensitivity test, establish that either coordinate is chaotic. Horizonwise analysis showed itself as good metric which helps us separate stability of the forecast of NSDE form. Small loss degradation metric itself showing us the ability of NSDE architectures determine noise and missing data in the input and make correct prediction.

References

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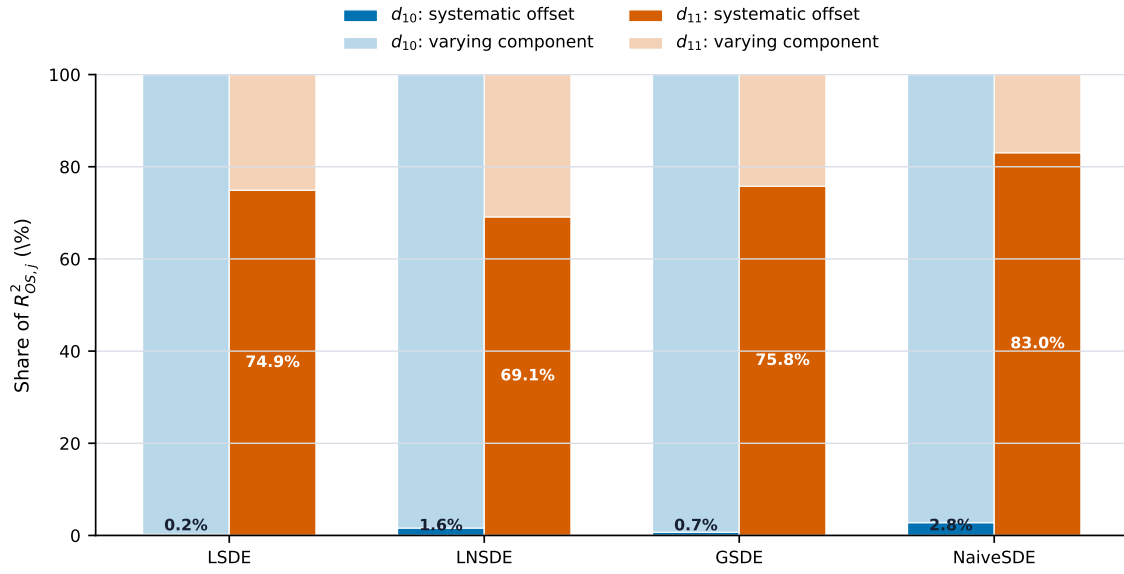


Figure 3: Bias decomposition of the pooled coordinate-wise ODE–SDE discrepancy. Channel 10 is dominated by a sign-changing component, whereas channel 11 is dominated by a persistent positive offset. Here $\text{Var}(\Delta_j)$ describes variation of ODE–SDE differences across examples, horizons, and conditions; it is not the variance across SDE Monte Carlo paths.

A Complete stress-test results

The column groups indicate missing rate; the three columns within each group indicate input-noise level. Overall RMS values are pooled through squared values rather than averaged arithmetically. Full-precision channel-10 data are also provided in the accompanying CSV files.

Table 3: Complete stress-test metrics for LSDE.

Metric	Missing 0.0			Missing 0.3			Missing 0.5			Missing 0.7			Overall
	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	
L_O	0.07761	0.07792	0.07873	0.07771	0.07808	0.07903	0.07784	0.07822	0.07931	0.07814	0.07852	0.07976	0.07841
L_S	0.04392	0.04395	0.04403	0.04392	0.04395	0.04404	0.04393	0.04397	0.04408	0.04395	0.04401	0.04417	0.04399
V_S	7.626e-04	7.626e-04	7.625e-04	7.626e-04	7.626e-04	7.625e-04	7.627e-04	7.627e-04	7.626e-04	7.628e-04	7.628e-04	7.625e-04	7.626e-04
Δ_O	-0.001305	0.002634	0.013028	0.000000	0.004651	0.016881	0.001654	0.006552	0.020478	0.005416	0.010319	0.026368	0.008890
Δ_S	-1.392e-05	7.290e-04	0.002517	0.000000	7.405e-04	0.002662	2.071e-04	0.001167	0.003592	6.075e-04	0.001967	0.005656	0.001653
R_{OS}	0.20190	0.20254	0.20441	0.20200	0.20269	0.20480	0.20212	0.20287	0.20530	0.20250	0.20337	0.20630	0.20341
R_{OY}	0.27859	0.27914	0.28058	0.27877	0.27942	0.28112	0.27900	0.27968	0.28161	0.27953	0.28021	0.28242	0.28001
R_{SY}	0.20957	0.20965	0.20984	0.20957	0.20965	0.20985	0.20959	0.20969	0.20995	0.20964	0.20978	0.21016	0.20975
$R_{OS,0}$	0.10022	0.10159	0.10538	0.10017	0.10148	0.10546	0.10017	0.10213	0.10714	0.10049	0.10294	0.10926	0.10308
$R_{OS,1}$	0.10940	0.10968	0.11069	0.10913	0.10935	0.11035	0.10859	0.10902	0.11030	0.10732	0.10801	0.10987	0.10931
$R_{OS,2}$	0.09689	0.09797	0.10164	0.09641	0.09765	0.10167	0.09600	0.09763	0.10249	0.09412	0.09626	0.10245	0.09847
$R_{OS,3}$	0.17418	0.17430	0.17479	0.17447	0.17491	0.17586	0.17492	0.17547	0.17667	0.17560	0.17610	0.17742	0.17539
$R_{OS,4}$	0.31571	0.31724	0.32231	0.31563	0.31693	0.32221	0.31540	0.31706	0.32349	0.31503	0.31801	0.32670	0.31883
$R_{OS,5}$	0.35042	0.35078	0.35171	0.35003	0.35044	0.35158	0.34938	0.34973	0.35096	0.34808	0.34834	0.34971	0.35010
$R_{OS,6}$	0.14513	0.14569	0.14777	0.14488	0.14566	0.14799	0.14460	0.14573	0.14861	0.14389	0.14495	0.14817	0.14610
$R_{OS,7}$	0.12584	0.12626	0.12743	0.12549	0.12615	0.12763	0.12499	0.12561	0.12718	0.12421	0.12499	0.12696	0.12607
$R_{OS,8}$	0.12236	0.12330	0.12623	0.12259	0.12362	0.12682	0.12290	0.12396	0.12750	0.12388	0.12499	0.12892	0.12477
$R_{OS,9}$	0.06538	0.06623	0.06888	0.06491	0.06590	0.06889	0.06432	0.06575	0.06958	0.06315	0.06509	0.07010	0.06655
$R_{OS,10}$	0.08516	0.08586	0.08780	0.08490	0.08553	0.08750	0.08463	0.08531	0.08755	0.08408	0.08498	0.08780	0.08593
$R_{OS,11}$	0.44962	0.45060	0.45266	0.45098	0.45201	0.45455	0.45277	0.45340	0.45593	0.45722	0.45738	0.45987	0.45392
$R_{OS,12}$	0.10674	0.10720	0.10830	0.10670	0.10714	0.10827	0.10666	0.10713	0.10836	0.10648	0.10695	0.10833	0.10736
$R_{OS,13}$	0.09751	0.09821	0.10025	0.09732	0.09839	0.10100	0.09731	0.09845	0.10138	0.09718	0.09826	0.10151	0.09891

Table 4: Complete stress-test metrics for LNSDE.

Metric	Missing 0.0			Missing 0.3			Missing 0.5			Missing 0.7			Overall
	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	
L_O	0.07761	0.07792	0.07873	0.07771	0.07808	0.07903	0.07784	0.07822	0.07931	0.07814	0.07852	0.07976	0.07841
L_S	0.05661	0.05665	0.05673	0.05658	0.05662	0.05670	0.05654	0.05657	0.05667	0.05643	0.05646	0.05659	0.05660
V_S	9.982e-04	9.982e-04	9.988e-04	9.981e-04	9.981e-04	9.987e-04	9.976e-04	9.979e-04	9.986e-04	9.967e-04	9.971e-04	9.980e-04	9.980e-04
Δ_O	-0.001305	0.002634	0.013028	0.000000	0.004651	0.016881	0.001654	0.006552	0.020478	0.005416	0.010319	0.026368	0.008890
Δ_S	6.141e-04	0.001257	0.002709	0.000000	6.218e-04	0.002136	-7.313e-04	-1.284e-04	0.001533	-0.002693	-0.002072	1.510e-04	2.830e-04
R_{OS}	0.19956	0.20026	0.20230	0.19966	0.20040	0.20270	0.19979	0.20065	0.20334	0.20022	0.20116	0.20435	0.20120
R_{OY}	0.27859	0.27914	0.28058	0.27877	0.27942	0.28112	0.27900	0.27968	0.28161	0.27953	0.28021	0.28242	0.28001
R_{SY}	0.23794	0.23801	0.23819	0.23787	0.23794	0.23812	0.23778	0.23785	0.23805	0.23754	0.23762	0.23788	0.23790
$R_{OS,0}$	0.14642	0.14739	0.15042	0.14626	0.14736	0.15068	0.14605	0.14788	0.15221	0.14605	0.14794	0.15298	0.14849
$R_{OS,1}$	0.09751	0.09807	0.09959	0.09702	0.09743	0.09883	0.09619	0.09662	0.09813	0.09429	0.09493	0.09698	0.09714
$R_{OS,2}$	0.14885	0.14945	0.15182	0.14833	0.14922	0.15194	0.14756	0.14885	0.15228	0.14589	0.14748	0.15182	0.14947
$R_{OS,3}$	0.17290	0.17316	0.17390	0.17301	0.17353	0.17467	0.17322	0.17383	0.17518	0.17360	0.17409	0.17554	0.17389
$R_{OS,4}$	0.24899	0.25130	0.25887	0.24922	0.25148	0.25964	0.24935	0.25220	0.26192	0.24972	0.25359	0.26559	0.25438
$R_{OS,5}$	0.31263	0.31300	0.31392	0.31226	0.31251	0.31349	0.31184	0.31207	0.31323	0.31101	0.31109	0.31239	0.31245
$R_{OS,6}$	0.13716	0.13774	0.13978	0.13708	0.13775	0.13995	0.13698	0.13790	0.14058	0.13690	0.13819	0.14167	0.13848
$R_{OS,7}$	0.15020	0.15059	0.15163	0.15013	0.15052	0.15162	0.15004	0.15046	0.15170	0.15012	0.15075	0.15244	0.15085
$R_{OS,8}$	0.13862	0.13976	0.14283	0.13877	0.13986	0.14305	0.13906	0.14022	0.14374	0.13983	0.14103	0.14491	0.14099
$R_{OS,9}$	0.13896	0.13939	0.14073	0.13866	0.13924	0.14079	0.13823	0.13910	0.14116	0.13737	0.13857	0.14137	0.13947
$R_{OS,10}$	0.11236	0.11291	0.11452	0.11216	0.11258	0.11416	0.11192	0.11235	0.11411	0.11153	0.11216	0.11437	0.11293
$R_{OS,11}$	0.42226	0.42324	0.42537	0.42342	0.42450	0.42719	0.42499	0.42580	0.42864	0.42905	0.42937	0.43222	0.42635
$R_{OS,12}$	0.11630	0.11647	0.11711	0.11671	0.11693	0.11771	0.11718	0.11748	0.11842	0.11809	0.11844	0.11964	0.11754
$R_{OS,13}$	0.17453	0.17493	0.17623	0.17443	0.17502	0.17668	0.17449	0.17515	0.17704	0.17416	0.17485	0.17704	0.17538

Table 5: Complete stress-test metrics for GSDE.

Metric	Missing 0.0			Missing 0.3			Missing 0.5			Missing 0.7			Overall
	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	
L_O	0.07761	0.07792	0.07873	0.07771	0.07808	0.07903	0.07784	0.07822	0.07931	0.07814	0.07852	0.07976	0.07841
L_S	0.04550	0.04555	0.04575	0.04539	0.04543	0.04563	0.04528	0.04533	0.04555	0.04495	0.04499	0.04523	0.04538
V_S	1.783e-04	1.784e-04	1.787e-04	1.785e-04	1.785e-04	1.789e-04	1.787e-04	1.786e-04	1.789e-04	1.792e-04	1.793e-04	1.796e-04	1.788e-04
Δ_O	-0.001305	0.002634	0.013028	0.000000	0.004651	0.016881	0.001654	0.006552	0.020478	0.005416	0.010319	0.026368	0.008890
Δ_S	0.002376	0.003379	0.007811	0.000000	8.514e-04	0.005122	-0.002581	-0.001514	0.003383	-0.009896	-0.008889	-0.003558	-2.931e-04
R_{OS}	0.19954	0.20039	0.20296	0.19958	0.20048	0.20330	0.19964	0.20064	0.20387	0.19976	0.20087	0.20463	0.20131
R_{OY}	0.27859	0.27914	0.28058	0.27877	0.27942	0.28112	0.27900	0.27968	0.28161	0.27953	0.28021	0.28242	0.28001
R_{SY}	0.21331	0.21342	0.21389	0.21306	0.21315	0.21361	0.21279	0.21290	0.21342	0.21200	0.21211	0.21268	0.21303
$R_{OS,0}$	0.12162	0.12268	0.12703	0.12136	0.12263	0.12724	0.12120	0.12326	0.12914	0.12075	0.12291	0.12967	0.12416
$R_{OS,1}$	0.08396	0.08483	0.08707	0.08375	0.08468	0.08698	0.08345	0.08427	0.08658	0.08220	0.08333	0.08630	0.08480
$R_{OS,2}$	0.13013	0.13130	0.13567	0.12944	0.13079	0.13543	0.12850	0.13020	0.13565	0.12587	0.12838	0.13528	0.13143
$R_{OS,3}$	0.21014	0.21068	0.21208	0.21023	0.21106	0.21288	0.21033	0.21117	0.21312	0.21064	0.21146	0.21356	0.21145
$R_{OS,4}$	0.29370	0.29592	0.30418	0.29318	0.29547	0.30429	0.29240	0.29562	0.30634	0.29055	0.29478	0.30767	0.29790
$R_{OS,5}$	0.29521	0.29575	0.29679	0.29476	0.29515	0.29628	0.29413	0.29452	0.29583	0.29276	0.29307	0.29462	0.29491
$R_{OS,6}$	0.10246	0.10359	0.10749	0.10202	0.10305	0.10687	0.10149	0.10301	0.10767	0.10036	0.10255	0.10842	0.10412
$R_{OS,7}$	0.15187	0.15291	0.15576	0.15179	0.15305	0.15606	0.15182	0.15291	0.15595	0.15195	0.15287	0.15595	0.15358
$R_{OS,8}$	0.09674	0.09812	0.10168	0.09676	0.09819	0.10206	0.09685	0.09834	0.10258	0.09697	0.09831	0.10286	0.09915
$R_{OS,9}$	0.10454	0.10502	0.10700	0.10406	0.10474	0.10699	0.10329	0.10430	0.10716	0.10207	0.10350	0.10729	0.10501
$R_{OS,10}$	0.11943	0.12006	0.12205	0.11907	0.11971	0.12180	0.11866	0.11941	0.12184	0.11764	0.11855	0.12136	0.11997
$R_{OS,11}$	0.46566	0.46657	0.46836	0.46709	0.46801	0.47026	0.46902	0.46942	0.47155	0.47344	0.47346	0.47569	0.46989
$R_{OS,12}$	0.09187	0.09249	0.09414	0.09185	0.09236	0.09394	0.09178	0.09239	0.09424	0.09165	0.09244	0.09463	0.09282
$R_{OS,13}$	0.08778	0.08858	0.09078	0.08761	0.08882	0.09168	0.08781	0.08902	0.09223	0.08729	0.08838	0.09196	0.08935

Table 6: Complete stress-test metrics for NaiveSDE.

Metric	Missing 0.0			Missing 0.3			Missing 0.5			Missing 0.7			Overall
	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	
L_O	0.07761	0.07792	0.07873	0.07771	0.07808	0.07903	0.07784	0.07822	0.07931	0.07814	0.07852	0.07976	0.07841
L_S	0.06217	0.06218	0.06221	0.06216	0.06217	0.06221	0.06215	0.06216	0.06220	0.06211	0.06213	0.06217	0.06217
V_S	1.771e-04	1.780e-04	1.801e-04	1.769e-04	1.778e-04	1.799e-04	1.765e-04	1.774e-04	1.795e-04	1.761e-04	1.769e-04	1.789e-04	1.779e-04
Δ_O	-0.001305	0.002634	0.013028	0.000000	0.004651	0.016881	0.001654	0.006552	0.020478	0.005416	0.010319	0.026368	0.008890
Δ_S	1.909e-04	3.690e-04	8.943e-04	0.000000	2.168e-04	7.778e-04	-1.874e-04	1.124e-04	7.624e-04	-6.965e-04	-4.382e-04	1.661e-04	1.806e-04
R_{OS}	0.18738	0.18801	0.19002	0.18755	0.18834	0.19073	0.18776	0.18859	0.19130	0.18825	0.18908	0.19223	0.18911
R_{OY}	0.27859	0.27914	0.28058	0.27877	0.27942	0.28112	0.27900	0.27968	0.28161	0.27953	0.28021	0.28242	0.28001
R_{SY}	0.24934	0.24936	0.24942	0.24931	0.24934	0.24941	0.24929	0.24933	0.24941	0.24923	0.24926	0.24933	0.24934
$R_{OS,0}$	0.11138	0.11267	0.11642	0.11159	0.11291	0.11693	0.11170	0.11313	0.11761	0.11238	0.11415	0.11960	0.11424
$R_{OS,1}$	0.10457	0.10484	0.10600	0.10456	0.10480	0.10595	0.10445	0.10470	0.10595	0.10388	0.10441	0.10614	0.10502
$R_{OS,2}$	0.08130	0.08260	0.08699	0.08050	0.08211	0.08698	0.07963	0.08157	0.08734	0.07677	0.07913	0.08641	0.08268
$R_{OS,3}$	0.17900	0.17926	0.18011	0.17913	0.17967	0.18094	0.17938	0.18001	0.18150	0.17963	0.18026	0.18196	0.18007
$R_{OS,4}$	0.25976	0.26156	0.26806	0.26019	0.26257	0.27027	0.26061	0.26365	0.27291	0.26153	0.26493	0.27586	0.26521
$R_{OS,5}$	0.30960	0.30997	0.31096	0.30940	0.30967	0.31074	0.30913	0.30923	0.31031	0.30860	0.30865	0.30992	0.30968
$R_{OS,6}$	0.12482	0.12535	0.12774	0.12451	0.12523	0.12785	0.12413	0.12522	0.12841	0.12318	0.12442	0.12823	0.12577
$R_{OS,7}$	0.11290	0.11292	0.11391	0.11241	0.11282	0.11426	0.11162	0.11205	0.11364	0.11024	0.11109	0.11332	0.11260
$R_{OS,8}$	0.09993	0.10116	0.10466	0.10000	0.10148	0.10545	0.10014	0.10167	0.10600	0.10050	0.10191	0.10656	0.10248
$R_{OS,9}$	0.08304	0.08373	0.08573	0.08287	0.08353	0.08563	0.08275	0.08364	0.08629	0.08205	0.08352	0.08727	0.08419
$R_{OS,10}$	0.06774	0.06843	0.07065	0.06767	0.06844	0.07086	0.06750	0.06836	0.07115	0.06756	0.06860	0.07190	0.06909
$R_{OS,11}$	0.44158	0.44238	0.44417	0.44281	0.44378	0.44620	0.44440	0.44500	0.44746	0.44825	0.44837	0.45083	0.44544
$R_{OS,12}$	0.09712	0.09753	0.09858	0.09740	0.09785	0.09901	0.09763	0.09808	0.09932	0.09836	0.09876	0.10009	0.09832
$R_{OS,13}$	0.10961	0.11017	0.11212	0.10928	0.11018	0.11269	0.10891	0.10985	0.11262	0.10757	0.10827	0.11117	0.11021

B Channel-10 values by stress condition

Table 7: Channel-10 RMS discrepancy across the 12 stress-test conditions.

Metric	Missing 0.0			Missing 0.3			Missing 0.5			Missing 0.7			Overall
	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	0.0	0.1	0.2	
LSDE	0.08516	0.08586	0.08780	0.08490	0.08553	0.08750	0.08463	0.08531	0.08755	0.08408	0.08498	0.08780	0.08593
LNSDE	0.11236	0.11291	0.11452	0.11216	0.11258	0.11416	0.11192	0.11235	0.11411	0.11153	0.11216	0.11437	0.11293
GSDE	0.11943	0.12006	0.12205	0.11907	0.11971	0.12180	0.11866	0.11941	0.12184	0.11764	0.11855	0.12136	0.11997
NaiveSDE	0.06774	0.06843	0.07065	0.06767	0.06844	0.07086	0.06750	0.06836	0.07115	0.06756	0.06860	0.07190	0.06909

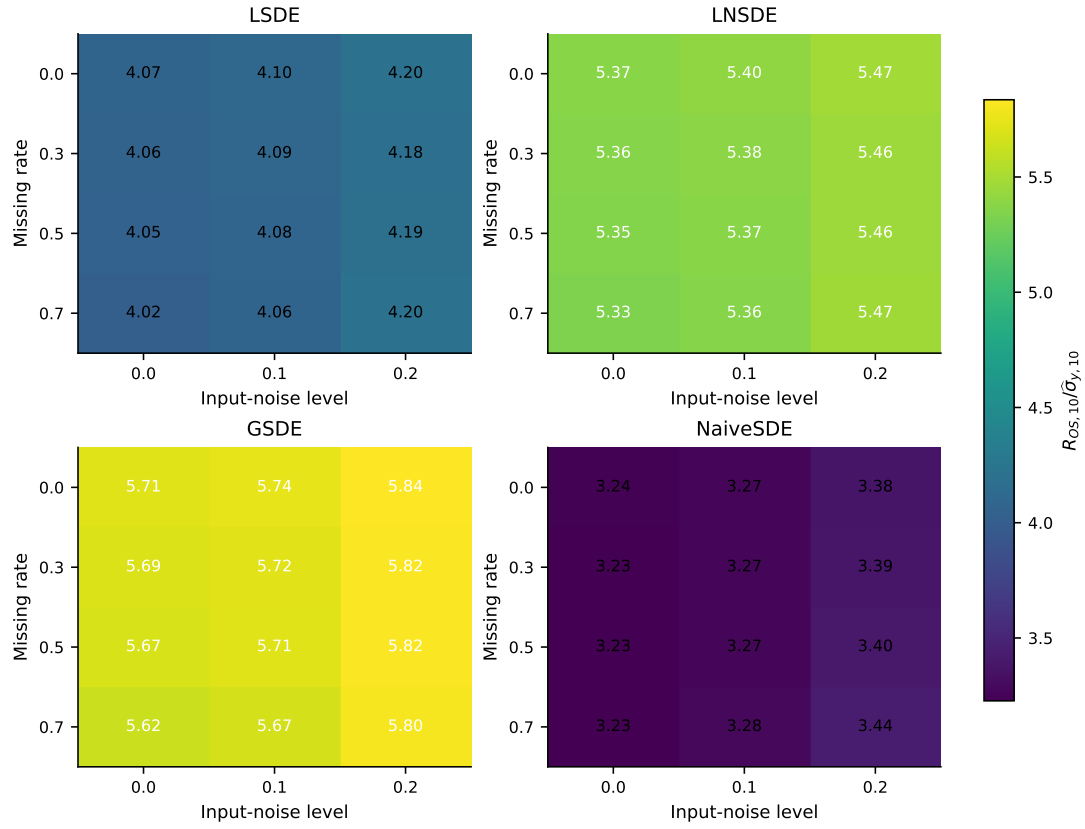


Figure 4: Scale-normalized channel-10 discrepancy over missing-rate and input-noise combinations. Input noise produces a clearer increase than missingness, although the model-to-model ordering is stable.